ECommerce API - Postman Documentation Index

Welcome to the complete Postman documentation for the ECommerce API! 📚

📁 Documentation Structure

```
postman/
├── collections/
│   └── ECommerce_API.json          ← Main API collection
├── environments/
│   └── Development.json            ← Environment variables
├── globals/
│   └── Global_Variables.json       ← Global constants
├── specs/
│   └── API_SPECIFICATION.md        ← Detailed API spec
├── README.md                       ← Full documentation
├── QUICK_START.md                  ← 5-minute setup
└── INDEX.md                        ← This file
```

🚀 Getting Started

New to the API? Start here:

1. **QUICK_START.md** - 5-minute setup
   - Import collection
   - Login & get token
   - Test basic endpoints
   - Common mistakes
2. **README.md** - Complete reference
   - Detailed workflows
   - All endpoints
   - Response formats
   - Troubleshooting
3. **API_SPECIFICATION.md** - Technical details
   - Data models
   - Business rules
   - Error handling
   - Database structure

📦 What's Included

Collection: `ECommerce_API.json`

Sections:

- 🔐 **Authentication** - Register, Login, Refresh Token, Forgot Password, Verify OTP, Reset Password
- 📦 **Products** - CRUD operations, filtering
- 🎨 **Product Variants** - Size/Color management
- 🛒 **Cart** - Shopping cart operations
- 📁 **Categories & Brands** - Master data
- 👨‍💼 **Admin Management** - Cart controls

Total Endpoints: 45+

Environment: `Development.json`

Key Variables:

- `baseUrl` - API base (localhost:5000/api)
- `authToken` - Auto-set after login
- `userId` - Auto-set after login
- Test IDs for quick testing

Globals: `Global_Variables.json`

Common Values:

- Test credentials
- Pagination defaults
- Test product/variant IDs

🔄 Popular Workflows

1️⃣ Create Product with Variants (Admin)

```
1. Create product → POST /api/products/create
2. Add variants → POST /api/products/{id}/variants (×3)
3. Stock auto-calculated → GET /api/products/{id}
```

See: [README.md - Product Management]

2️⃣ Customer Shopping

```
1. Login → POST /api/auth/login
2. Browse → GET /api/products
3. Add to cart → POST /api/cart/items
4. View cart → GET /api/cart/summary
```

See: QUICK_START.md - Customer Flow

3 Admin Stock Management

```
1. Update variant stock → PUT /api/products/{id}/variants/{variantId}
2. Product stock auto-recalculated
3. View all carts → GET /api/admin/carts
```

See: README.md - Admin Endpoints

🎱 Key Business Rules

Two Types of Products

WITH Variants (e.g., T-Shirt with sizes)

```
HasVariants = true
→ Must select variant when adding to cart
→ Stock managed at variant level
→ Product stock = SUM of variant stocks (auto-calculated)
```

WITHOUT Variants (e.g., Poster)

```
HasVariants = false
→ Add directly to cart
→ Stock managed at product level
```

See: API Specification - Business Rules

📊 API Response Format

All endpoints follow this structure:

Success (200 OK)

```
{
  "status": true,
  "message": "Operation successful",
  "data": { /* response data */ }
}
```

Error (4xx/5xx)

```
{
  "status": false,
  "message": "Error description",
  "data": null
}
```

See: README.md - Response Format

🔗 Quick Links by Role

For Frontend Developers

- Product browsing: README.md - Product Management
- Cart operations: README.md - Shopping Cart Logic
- Common workflows: QUICK_START.md

For Backend Developers

- API architecture: API_SPECIFICATION.md
- Data models: API_SPECIFICATION.md - Data Model
- Error handling: API_SPECIFICATION.md - Error Handling

For QA/Testers

- Test workflows: README.md - Common Workflows
- Error scenarios: README.md - Troubleshooting
- All endpoints: Collection ECommerce_API.json

For Admins

- Admin endpoints: README.md - Admin Endpoints
- Stock management: API_SPECIFICATION.md - Stock Management

⚡ Common Tasks

Task	Resource	Time
Get started quickly	QUICK_START.md	5 min
Understand stock system	API_SPECIFICATION.md	10 min
Find specific endpoint	README.md	2 min
Test cart flow	Collection → Cart section	5 min
Debug error	README.md - Troubleshooting	3 min

Task	Resource	Time
Setup environment	README.md - Getting Started	2 min

📞 Support Resources

Documentation Files

- **QUICK_START.md** - Fast, practical guide
- **README.md** - Comprehensive reference
- **API_SPECIFICATION.md** - Technical details
- **This file** - Navigation & overview

Collection Features

- 40+ pre-built requests
- Auto-token capture from login
- Environment variables for easy switching
- Test scripts for validation

Getting Help

1. Check Troubleshooting section
2. Review relevant workflow in Common Workflows
3. Check API specification for business rules
4. Look for similar endpoint in collection

📋 File Checklist

Before using the API, ensure you have:

- ☑ `ECommerce_API.json` - Main collection
- ☑ `Development.json` - Development environment
- ☑ `Global_Variables.json` - Test constants
- ☑ `README.md` - Full documentation
- ☑ `QUICK_START.md` - Setup guide
- ☑ `API_SPECIFICATION.md` - Technical reference

🔄 Environment Setup

Development Environment

```
{
  "baseUrl": "http://localhost:5000/api",
  "adminBaseUrl": "http://localhost:5000/api/admin",
  "authToken": "Bearer ...",      // Auto-filled
```

```
  "userId": "...",                // Auto-filled
  "page": "1",
  "pageSize": "10"
}
```

Variables Auto-Set After Login

- `authToken` ← Used for all authenticated requests
- `userId` ← Your user ID

🎯 Next Steps

1. **Import collection** → Collections/ECommerce_API.json
2. **Set environment** → Environments/Development.json
3. **Read QUICK_START** → QUICK_START.md
4. **Login** → POST /api/auth/login
5. **Test endpoints** → Start with Products, then Cart

📝 Last Updated

Date: March 1, 2026
API Version: 1.0.0
Framework: ASP.NET Core 10.0

📚 Document Hierarchy

```
BEGIN HERE
    ↓
QUICK_START.md (5 min overview)
    ↓
README.md (move to specific section)
    ├→ Product Management?
    ├→ Shopping Cart?
    ├→ Admin Endpoints?
    └→ Troubleshooting?
    ↓
API_SPECIFICATION.md (technical deep dive)
    ├→ Data Models
    ├→ Business Rules
    ├→ Stock System
    └→ Database Schema
    ↓
Collection (hands-on testing)
```

Happy coding! 🚀